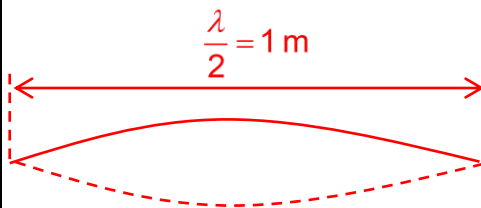


17	A standing wave is set up on a rope of length 1.0 m fixed at both ends. Which statement is correct?
A	The fundamental mode has a wavelength of 1 m.
B	The midpoint of the rope is always stationary.
C	There are two nodes on the rope when the fundamental frequency is doubled.
D	A standing wave of wavelength 0.4 m may be set up on the rope.

Answer: D

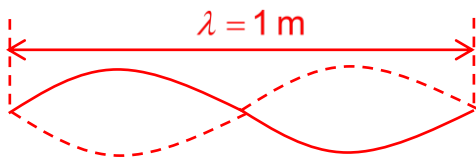


At the fundamental mode,

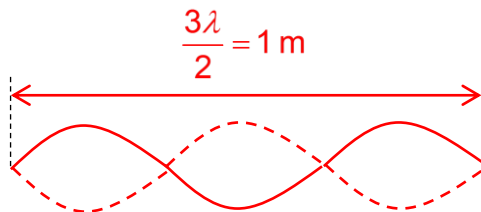
$\frac{\lambda}{2} = 1 \text{ m}$, $\lambda = 2 \text{ m}$ → Option A is incorrect.

The mid-point of the rope is not stationary at fundamental mode. Option B is incorrect.

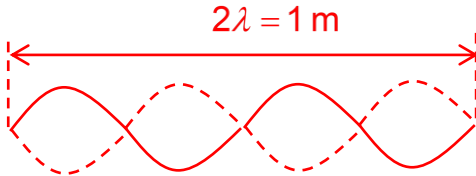
2nd Harmonic- $\lambda = 1 \text{ m}$



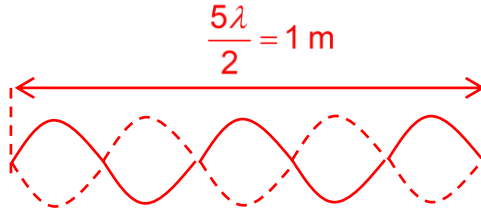
3rd Harmonic- $\lambda = 0.667 \text{ m}$



4th Harmonic- $\lambda = 0.5 \text{ m}$



5th Harmonic- $\lambda = 0.4 \text{ m}$



When the fundamental frequency is doubled, there are 3 nodes on the rope. Option C is incorrect.

At the 5th harmonic, the wavelength is 0.4 m. Option D is correct.